

AMENDMENT NO. 3 FEBRUARY 2009
TO
IS 10243 : 1993 2, 4-D ETHYL ESTER EC — SPECIFICATION

(*First Revision*)

(Page 1, clause 3.5.1, line 1) — Substitute ‘Either Annex A or Annex B shall be used. However, in case of dispute, Annex B shall be the referee method’ for ‘When determined by the method prescribed in Annex A’.

(Page 3, Annex A, clause A-4.2) — Add the following new Annex after A-4.2:

ANNEX B
(*Clause 3.5.1*)

DETERMINATION OF 2, 4-D ETHYL ESTER CONTENT

B-1 PRINCIPLE

2, 4-D ethyl ester content is determined by gas chromatography, using internal standard technique.

B-2 APPARATUS

B-2.1 Gas Liquid Chromatograph, gas chromatograph (GLC) equipped with FID and coupled to a printer-plotter-cum-integrator is used for this determination. The operative conditions suggested below are typical, which can be changed provided standardization is done:

Column	Glass, length = 200 cm, I.D. = 6 mm, packed with 3 percent OV-17 on Chromosorb WHP, 80-100 mesh
Temperatures	: Column — 180°C Injector — 250°C Detector — 260°C
Carrier gas	: Nitrogen 30 ml/min
Hydrogen	: 25 ml/min
Air	: 300 ml/min

B-2.2 Micro Syringe, 2- μ l capacity.

B-2.3 Standard Glassware

B-3 REAGENTS

B-3.1 Acetone, analytical reagent (AR) grade.

B-3.2 Internal Standard — Di-*n*-Butyl Phthalate, AR grade or equivalent.

B-3.3 2, 4-D Ethyl Ester Reference Standard, of known purity.

B-4 PROCEDURE

B-4.1 Preparation of Internal Standard Solution

Weigh out accurately 2.5 g of di-*n*-butyl phthalate into a 500-ml volumetric flask. Dissolve in acetone and make up the volume up to the mark with acetone. Shake well to homogenize.

B-4.2 Preparation of Standard Solution

Weigh out accurately 0.2 g of 2, 4-D ethyl ester reference standard of known purity into a 25-ml volumetric flask. Add 20-ml of the internal standard solution and make up the volume up to the mark with acetone. Shake well to homogenize.

B-4.3 Preparation of Sample Solution

Weigh out accurately quantity of a sample equivalent to 0.2 g of 2, 4-D ethyl ester into a 25-ml volumetric flask. Add 20 ml of the internal standard solution and make up the volume up to the mark with acetone. Shake well to homogenize.

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B-4.4 Estimation

B-4.4.1 Inject 1 µl of the reference standard solution until the internal standard/reference substance area quotients of two successive chromatograms do not deviate from each other by more than 2 percent. Then use the following injection sequence:

.....C S₁ S₁, C S₂ S₂, C S₃ S₃,

where

C = standard solution; and

S = sample solution (1, 2, 3 . . . n)

From the chromatograms of the standard solution and sample solution measure the peak areas of internal standard and 2, 4-D ethyl ester peaks and compute the percentage of the 2, 4-D ethyl ester as given in **B-5**.

B-4.4.2 Retention Time (Guide Values)

2, 4-D Ethyl ester	:	3.03 min (approx)
Di-n-butyl phthalate	:	7.03 min (approx)

A typical chromatogram is given in Fig. 1

FIG. 1 A TYPICAL CHROMATOGRAM

B-5 CALCULATION

$$\text{2, 4-D ethyl ester content, percent by mass} = \frac{M_1 \times A_3 \times A_2 \times P}{M_2 \times A_4 \times A_1}$$

where

- M₁ = mass in g of standard 2, 4-D ethyl ester in standard solution;
- M₂ = mass in g of sample taken for test;
- A₁ = peak area of 2, 4-D ethyl ester in the chromatogram of standard solution;
- A₂ = peak area of 2, 4-D ethyl ester in the chromatogram of sample solution;
- A₃ = peak area of internal standard in the chromatogram of standard solution;
- A₄ = peak area of internal standard in the chromatogram of sample solution; and
- P = percentage purity of 2, 4-D ethyl ester reference standard.

(FAD 1)